# Lecture 3: Relationships between variables and hypothesis tests

## Empirical Methods for Law and Economics | EDLE

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# Today's questions

Do more judges make courts faster?

Does more police mean less crime?

- By the end of today you will be able to:
  - describe a relationship between two variables with a scatter plot and a correlation;
  - give three reasons why an association may not be a causal effect;
  - test a difference in means and a difference in proportions, and read the p-value correctly.

## DISCUSS Back to "A Fine Is a Price"

- Last week you objected: the day-care centres are private, so parents may treat a fine as a price more easily.
- Look again at Figure 1. Some centres received the fine, others did not; all are observed before the fine, while it is in force, and after it is removed.

**Questions:**

1. Which **four** averages would you compare to measure the effect of the fine?
2. Does the private-centre objection threaten the comparison *within* the study, or the extension of the result to other settings?
3. What is the real number of independent units?

## ANSWERS Back to "A Fine Is a Price"

- Four averages: fined and control centres, before and during the fine. Effect = (fined: during − before) − (control: during − before). The design is a **difference-in-differences**.
- The private-centre objection concerns **external** validity (other parents, public centres, larger fines), not the comparison within the study.
- Independent units: **10 centres** (6 fined, chosen randomly; 4 controls), not 200 centre-weeks.
- One level of the fine says little about magnitudes, but one counterexample is enough against "a fine always deters".

# Internal and external validity

**Internal validity**

- Is the comparison credible *in the units studied*?
- Threats: groups that differ before treatment, other events at the same time, non-independent observations.

**External validity**

- Would the result hold *elsewhere*: other people, places, sizes of the intervention?
- Threats: selected samples (private centres), a single level of treatment, a specific period.

- Both matter; they fail for different reasons. A referee report should say which one a comment concerns.
- Today we work with country data: the question will mostly be whether a correlation can be read as an effect at all.

## PREDICT More judges, faster courts?

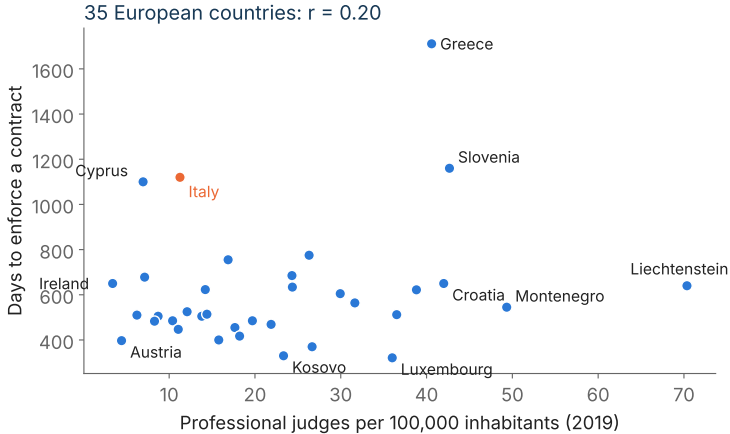
- New data: Eurostat, *Personnel in the criminal justice system* and *Police-recorded offences*, 2019, 37 European countries, merged with the Doing Business data you know.
- One row = one **country**. Judges and police are counted per 100,000 inhabitants.

**Write down before you see the data:**

1. The sign of the relationship between judges per 100,000 inhabitants and days to enforce a contract.
2. One country you expect to be an exception.

Sources: Eurostat (crim_just_job, crim_off_cat), 2019, free reuse with source acknowledgement; World Bank Doing Business 2020 (CC BY 4.0). Selected and merged by the instructor.

# Two variables: the scatter plot




Germany and Switzerland: number of judges not reported for 2019. Time: Doing Business 2020 standardized case, largest business city.

## Correlation: one number for a linear pattern

$$r_{xy} = \frac{1}{n-1} \sum_{i=1}^{n} \left( \frac{x_i - \bar{x}}{s_x} \right) \left( \frac{y_i - \bar{y}}{s_y} \right)$$

- Between $-1$ and $+1$; no units; the same if we swap $x$ and $y$.
- Measures **linear** association only. A strong curve can have $r$ close to 0.
- Sensitive to extreme values: one point far from the others can create or hide a correlation.
- Here: $r = 0.20$ ($n = 35$). Weak, and positive: more judges go with slightly *slower* enforcement.

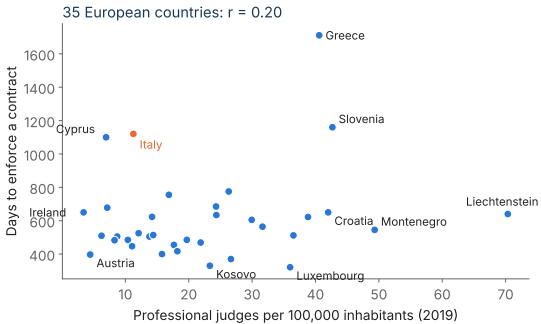
## STATA Guided Stata 1: scatter and correlation

```
use "$course_data/justice_europe_2019.dta", clear
describe
scatter time_days judges_100k, mlabel(country)
pwcorr time_days judges_100k, sig obs
```

| | |
|---|---|
| Correlation of `time_days` and `judges_100k` | 0.196 |
| p-value ($H_0$: correlation is zero) | 0.260 |
| Observations | 35 |

- `pwcorr` uses all rows where **both** variables are non-missing: 35, not 37.

# DISCUSS More judges, slower courts: should Greece fire judges?




## Questions:

1. Give one story in which slow courts *cause* more judges.
2. Give one third factor that could raise both.
3. Is a "professional judge" the same thing in every country?
4. What does a 35-country correlation tell a ministry of justice?

## ANSWERS More judges, slower courts?

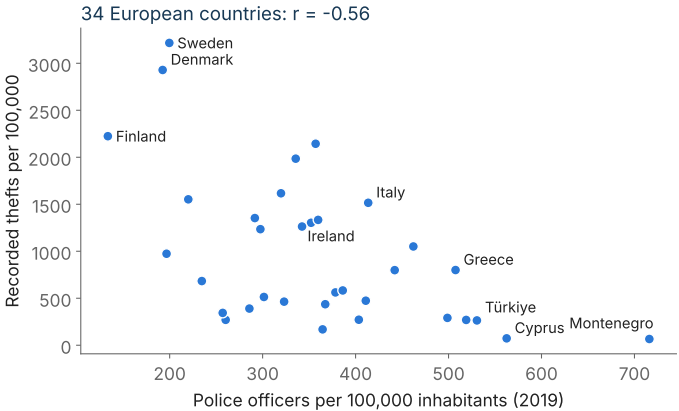
- **Reverse causality**: governments hire judges where backlogs are long.
- **Confounding**: legal tradition and procedure, volume of litigation.
- **Measurement**: a "professional judge" differs across systems (career judiciaries, lay magistrates); Doing Business times one standardized case in the largest business city.
- A correlation of 0.20 across 35 countries ($p = 0.26$) cannot tell a ministry what hiring judges would do.

# Association is not causation: three alternative explanations

| | Question to ask | Today's example |
|---|---|---|
| Reverse causality | Could $y$ cause $x$? | Backlogs lead governments to hire judges. |
| Confounding | Is there a third variable behind both? | Income drives both police staffing and the reporting of theft. |
| Measurement | Does the variable measure what its name says? | "Recorded" crime counts what is reported and recorded. |

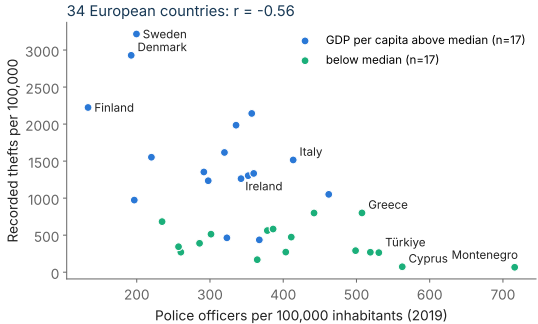
- Chance is a fourth: with few countries, a correlation can appear by accident. The test after the break addresses only this one.

# DISCUSS More police, less theft?



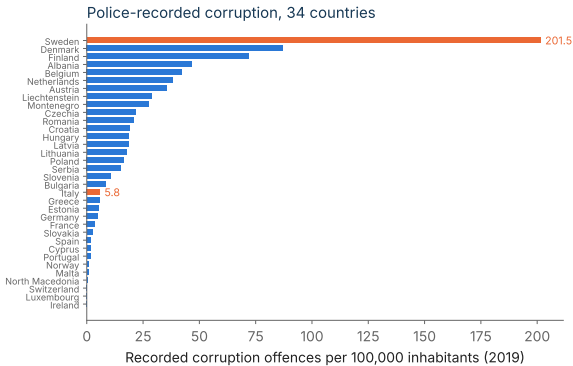

**Does policing deter theft? What else could produce this picture?**

# ANSWERS More police, less theft?




- A theft is *recorded* only if it is reported: reporting is higher where insurance requires it and trust in the police is high.
- Richer countries record more theft ($r = 0.64$, $n = 36$) and have fewer police officers ($r = -0.54$, $n = 34$): income is a candidate confounder, acting partly through measurement.
- Deterrence cannot be read from this picture.

# DISCUSS Is Sweden the most corrupt country in Europe?




- Recorded corruption offences per 100,000 inhabitants, 2019: Sweden 201.5, Denmark 87.2, Finland 71.8; Italy 5.8; Greece 5.7.

**What does this variable measure?**

- Corruption that occurs $\times$ the share that is detected $\times$ how offences are counted.

## ANSWERS Is Sweden the most corrupt country in Europe?

- Recorded corruption = offences committed × share detected and recorded × national legal definitions.
- Active anti-corruption enforcement and broad definitions *raise* the count: the Nordic countries record more.
- A variable produced by the legal system cannot be used naively to evaluate that system.

## PREDICT After the break: are richer economies faster?

- Back to all 191 economies of Doing Business. Split them into high income (World Bank classification) and the rest.

| Group | Economies | Mean days | Median days |
|---|---|---|---|
| Not high income | 132 | 662.4 | 590.5 |
| High income | 59 | 621.3 | 545.0 |

**Bet before we test:**

- Is a gap of 41 days large? Would it survive if the World Bank had measured a different set of economies, or a different year?

# The logic of a test

1. **Null hypothesis** $H_0$: the two population means are equal. Alternative: they differ.
2. **Test statistic**: signal divided by noise, the same standard error as in Lecture 2:

$$t = \frac{\bar{y}_0 - \bar{y}_1}{\mathrm{SE}(\bar{y}_0 - \bar{y}_1)}$$

3. **p-value**: if $H_0$ were true, the probability of a $t$ at least as far from 0 as the one we observed.

- A small p-value: the data would be surprising if $H_0$ were true.
- The p-value is **not** the probability that $H_0$ is true, and not the size or importance of the difference.

## STATA Guided Stata 2: a difference in means

```
use "$course_data/contracts_2019.dta", clear
generate byte high_income = (income_group == "High income")
ttest time_days, by(high_income)
```

| | |
|---|---|
| diff = mean(not high) − mean(high) | 41.2 days |
| Standard error | 47.7 |
| $t$ (189 degrees of freedom) | 0.86 |
| p-value, two-sided | 0.389 |
| 95% confidence interval for the difference | −52.9 to 135.2 days |

- With `, unequal` (different variances): $t = 0.89$, $p = 0.373$. Same conclusion.

## DISCUSS "p = 0.39: income does not matter for courts"

**What is wrong with this sentence?**

1. Absence of evidence is not evidence of absence: what does the confidence interval allow?
2. These 191 economies are almost all the economies in the world. What "population" is the test about?
3. The high-income group mixes very different economies. Is a two-group split the right comparison?

## ANSWERS "p = 0.39: income does not matter"

- Not significant $\neq$ no difference: the 95% interval, $-53$ to 135 days, includes zero and also differences of more than four months.
- With 191 of 191 economies, the test treats the 2019 values as one draw from a process (other years, other expert answers, measurement error). A standard modelling choice, to be stated.
- Two groups discard information: income is continuous. Lecture 4 uses GDP per capita.
- Correct sentence: "We cannot conclude that income matters, and we cannot conclude that it does not."

## STATA Guided Stata 3: a difference in proportions

- Outcome: does the court system have *any* automation feature (`court_automation` $> 0$)?

```
generate byte any_automation = (court_automation > 0) ///
    if !missing(court_automation)
tabulate high_income any_automation, row
prtest any_automation, by(high_income)
```

| | |
|---|---|
| Share with automation: not high income | 47.7% (63 of 132) |
| Share with automation: high income | 93.2% (55 of 59) |
| diff = prop(not high) − prop(high) | −0.455 |
| $z$ | −5.98 |
| p-value, two-sided | $< 0.001$ |
| 95% confidence interval | −0.56 to −0.35 |

# Two kinds of error, and many tests

- **Type I**: rejecting a true $H_0$. At the 5% level this happens in 5% of tests where $H_0$ is true.
- **Type II**: not rejecting a false $H_0$. Likely when the sample is small or the effect is small (low power).

- Police against six crime categories: four correlations have $p < 0.05$, all negative.
- Twenty independent tests of true nulls: on average one "significant" result by chance.
- Here the four are not independent chance findings: one confounder (income, reporting) can produce all of them.

## Laboratory

- Worksheet: `do/exercises/03_relationships_and_tests.do`. Lab Companion, Lab 3.
- Data: `justice_europe_2019.dta` (Tasks 1–4) and `contracts_2019.dta` (Tasks 5–7).
- For every number: unit, number of observations, and one sentence that a lawyer could understand.

# Exit check

1. The correlation between judges and enforcement time is 0.20. Give one reason why hiring judges might still shorten trials.
2. Recorded thefts are higher in Sweden than in Montenegro. Name one reason that is not “there is more theft in Sweden”.
3. A test gives $p = 0.39$ and a 95% interval of $-53$ to 135 days. Write one correct sentence about it.

## For the next meeting

- Read Voigt and Wulf (2019), "What makes prosecutors independent?", *Journal of Institutional Economics*: abstract, Sections 1 and 5, Section 7 and the first two paragraphs of Section 8 (pp. 112–113), Table 1 and column (1) of Table 2.
- Bring answers to three questions:
  1. What is one observation in their data?
  2. How do they measure prosecutorial independence, and is it *de jure* (law on paper) or *de facto* (practice)?
  3. Pick one coefficient in column (1) of Table 2: what does it say, in words, with its unit?
- Next meeting: Italy takes 1,120 days. Slow compared to what?